

Management of Malignant Pleural Effusion

A Cost-Utility Analysis

Majid Shafiq, MD, MPH,* Kevin D. Frick, PhD,† Hans Lee, MD,‡
Lonny Yarmus, MD,‡ and David J. Feller-Kopman, MD,‡

Background: Malignant pleural effusion (MPE) is associated with a significant impact on health-related quality of life. Palliative interventions abound, with varying costs and degrees of invasiveness. We examined the relative cost-utility of 5 therapeutic alternatives for MPE among adults.

Methods: Original studies investigating the management of MPE were extensively researched, and the most robust and current data particularly those from the TIME2 trial were chosen to estimate event probabilities. Medicare data were used for cost estimation. Utility estimates were adapted from 2 original studies and kept consistent with prior estimations. The decision tree model was based on clinical guidelines and authors' consensus opinion. Primary outcome of interest was the incremental cost-effectiveness ratio for each intervention over a less effective alternative over an analytical horizon of 6 months. Given the paucity of data on rapid pleurodesis protocol, a sensitivity analysis was conducted to address the uncertainty surrounding its efficacy in terms of achieving long-term pleurodesis.

Results: Except for repeated thoracentesis (RT; least effective), all interventions had similar effectiveness. Tunneled pleural catheter was the most cost-effective option with an incremental cost-effectiveness ratio of \$45,747 per QALY gained over RT, assuming a willingness-to-pay threshold of \$100,000/QALY. Multivariate sensitivity analysis showed that rapid pleurodesis protocol remained cost-ineffective even with an estimated probability of lasting pleurodesis up to 85%.

Conclusions: Tunneled pleural catheter is the most cost-effective therapeutic alternative to RT. This, together with its relative convenience (requiring neither hospitalization nor thoroscopic procedural skills), makes it an intervention of choice for MPE.

Key Words: pleural diseases, pleural effusion, malignant, cost-effectiveness

(*J Bronchol Intervent Pulmonol* 2015;22:215–225)

Malignant pleural effusion (MPE) is defined as the presence of pleural effusion secondary to cancerous pleural involvement, most commonly occurring alongside lung and breast cancer.^{1–3} The worldwide disease burden is unknown but the estimated annual incidence in United States alone is over 150,000.¹ According to 1 postmortem case series, 15% of patients dying from any malignancy had MPE.⁴ MPE carries a poor prognosis; variable estimates of survival have been published but the most robust and prospectively gathered data point to approximately 6 months.^{3,5–7} Almost all patients are symptomatic, dyspnea and chest pain being the most common symptoms, and the disease has a sizeable impact on health-related quality of life (HRQOL).²

Beyond opioids and supplemental oxygen, various palliative interventions are available, including repeated thoracentesis (RT), chemical pleurodesis typically using talc by either thoroscopic talc poudrage (TP) or chest tube-guided talc slurry (TS), tunneled pleural catheter (TPC), and a rapid pleurodesis protocol (RPP) that involves TP via medical thoracoscopy along with concurrent TPC placement.^{2,8} These therapies offer unique advantages, disadvantages, costs, and degrees of invasiveness, while producing a measurable improvement in patient-reported outcomes such as dyspnea, chest pain, and HRQOL.^{2,6,9} For each approach, the health care utilization costs can be compared with HRQOL gains to understand tradeoffs that patients, health care providers, and institutions, and third-party payers must make.

Received for publication February 14, 2015; accepted May 21, 2015.

From the *Division of Pulmonary and Critical Care Medicine, Stanford University School of Medicine, Stanford, CA; †Department of Health Policy and Management, Johns Hopkins Bloomberg School of Public Health; and ‡Division of Pulmonary and Critical Care Medicine, Johns Hopkins University School of Medicine, Baltimore, MD.

Presented at the American Thoracic Society 2015 International Conference on May 18, 2015 in Denver, CO.

Disclosure: D.J.F.-K. has received consulting fees from CareFusion Inc. For the remaining authors there is no conflict of interest or other disclosures.

Reprints: Majid Shafiq, MD, MPH, Stanford University School of Medicine, 300 Pasteur Drive, H3143, Stanford, CA 94305-5236 (e-mail: shafiq@stanford.edu).

Supplemental Digital Content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's Website, www.bronchology.com.

Copyright © 2015 Wolters Kluwer Health, Inc. All rights reserved.

Olden et al¹⁰ conducted a cost-effectiveness comparison between TS and TPC and reported that TS dominated TPC for an analytical horizon of 6 months (TS provided 0.06 additional QALYs at a cost lower by nearly \$1000). Puri and colleagues published a second analysis in which RT and TP were added to the list of competing alternatives. TPC was the dominant approach for an analytical horizon of 3 months, but TS became dominant for a horizon of 1 year.¹¹ Both analyses estimated costs from a third-party payer's perspective. Neither included RPP. These analyses predated the landmark TIME2 randomized trial published in 2012, which compared clinical and patient-reported outcomes between the only 2 strategies common to these analyses: TS and TPC.⁶ We sought to perform a cost-utility analysis of all aforementioned interventions using current and thorough cost estimates along with current and robust clinical efficacy and health utility data, including data from the TIME2 trial.

METHODS

The target population comprised of adults above 18 years of age with clinically or histopathologically established MPE from any cancer type who had undergone therapeutic thoracentesis with no evidence to suggest lung entrapment. Patients with a prior history of pleurodesis or TPC placement, or those with existing evidence of a nonexpandable lung, were excluded.

As the decision model was populated by aggregate data from published sources, and no patient records were used, institutional review board approval was not required.

Competing Alternatives

- (1) RT: An ultrasound-guided therapeutic thoracentesis is performed in the outpatient setting. It typically needs to be repeated on a monthly basis.^{12–14} The most common complication is a pneumothorax, which can be significant enough to require hospitalization and chest tube placement in approximately 1% of cases.
- (2) TS: The patient is hospitalized and an ultrasound-guided chest tube is inserted, draining the effusion and subsequently instilling TS into the pleural space. The intervention requires on average 4 days of hospitalization as the chest tube needs to stay in place before resolution of drainage (indicating successful pleurodesis).^{3,6} Ungraded/mixed talc (as opposed to graded talc with particle size > 15 mm only), which is commonly used in the United States, has been associated with acute talc pneumonitis (that could present as ARDS) whether administered as TS or TP. This complication occurs in 2% to 8% of cases, typically within the first 48 to 72 hours.^{6,15–17} This complication would prolong the index hospitalization but typically not lead to a rehospitalization. It is unlikely for chemical pleurodesis to result in a later complication needing rehospitalization, although rehospitalization could be needed in the context of pleurodesis failure requiring a second intervention.^{6,9}
- (3) TP: The patient is hospitalized and undergoes medical thoracoscopy, drainage of the effusion and spraying (poudrage) of aerosolized talc powder into the pleural space. This procedure is believed to have similar efficacy and complication rates as TS.^{2,3}
- (4) TPC: An ultrasound-guided TPC is placed in the outpatient setting.^{6,18} The TPC is left in place to allow intermittent drainage, requiring approximately 3 drainage bottles per week. Either a visiting nurse or a family member performs catheter care. Nearly half of the patients end up achieving pleurodesis, allowing removal of the TPC. In Putnam and colleagues's landmark study from the 1990s, this occurred at a median of 1 month into TPC placement, although more recent studies including Tremblay and colleagues and TIME2 trial report medians close to 2 months.^{6,9,19} Serious pleural infections are the most common late complication requiring rehospitalization. In the largest observational study examining TPC infection rates, empyema occurred in 4.9% of patients.²⁰ According to data from the TIME2 trial, approximately 10% of TPC recipients ended up getting hospitalized for intravenous antibiotic administration.⁶ All of them developed this complication well over 1 month into TPC placement, none required removal of the TPC, and approximately half of them went on to achieve spontaneous pleurodesis just like others.
- (5) RPP: The patient undergoes TP with simultaneous placement of a TPC followed by a 24-hour hospitalization.⁸ According to the limited observational data currently available, TPC is successfully removed after demonstration of pleurodesis in over 90% of patients at a median of 1 week after placement. It is not known what proportion

of patients experience a recurrence of the effusion after the initial pleurodesis. Arguably, these rates should not be dissimilar to those seen with TP alone.

Study Design

Base-case Analysis

A base-case analysis was carried out using a decision tree model approach (Tree Age Pro Healthcare 2014 software). The decision tree model constructed for this analysis is included in the online Appendix A, Supplemental Digital Content 1, <http://links.lww.com/LBR/A120>. We also studied each of the 2 decision tree models used in previously published cost-effectiveness studies after formulating our own model.^{10,11} Careful consideration was given to each point of difference between our model and each of the other 2 models in an effort to make improvements in our own model and stave off any inadvertent pitfalls and deficiencies.

Sensitivity Analysis

In view of the limited data available for RPP, a multivariate (nonrandom) sensitivity analysis was carried out to examine the effect of variation in its estimated incidence of effusion recurrence (temporary RPP success). As RPP can be reasonably expected to achieve lasting pleurodesis (permanent RPP success) at least as frequently as TP alone, the analysis was performed in a unidirectional manner and the probability of permanent RPP success was increased in 5% increments from 70% up to 85%. The estimated probability of pleurodesis failure (necessitating a chronically indwelling TPC) was kept constant.

Cost-effectiveness (also termed cost-utility as health utilities were used to calculate

effectiveness) was defined as the incremental cost-effectiveness ratio (ICER), that is, additional cost per QALY gained for each intervention as compared with a less costly one.

Analytical Horizon

We adopted an analytical horizon of 6 months.²¹ We have detailed our rationale for doing so in the discussion.

Perspective

Like previous analyses, this analysis was conducted from a third-party payer perspective and the United States' Medicare data were used for cost estimation. The assumption underlying this perspective is that the third-party payer shares the objective of maximizing population QALYs as long as the costs are not prohibitive. For obvious reasons, this assumption is more likely to hold true for public payers such as Medicare. Medicare is also the obvious choice because of its near-universality as an available payer for most hospitals across the country, its status as the biggest contributor to public health care expenditure, and the fact that many insurance payers often adopt Medicare charges or otherwise use it as a reference standard.²² Arguably of greatest relevance is the fact that most adults with MPE are 65 years old or older and, hence, Medicare eligible.^{6,22}

Outcomes

Table 1 lists the probability estimates for each outcome. Notably, authors did not pool probability data from all available studies but instead chose to adopt the most recent RCT data where possible. Hence, constancy of effect was not assumed. This meant that event rates from the TIME2 RCT were used for both TPC and TS.⁶ Event rates for TS were also extrapolated to TP as these are widely considered to be largely

TABLE 1. Event Probability Estimates for Each Intervention

Intervention	Probability			
	Pleurodesis Achieved Throughout 6 mo	Pleurodesis Not Achieved Throughout 6 mo (Months With Pleurodesis)	Complication Leading to New Hospital Admission	Failure at Outset Warranting Another Intervention
Repeated thoracentesis	0.00	1.0 (0)	0.01*	0.00
Talc slurry	0.67	0.22 (2)	0.00	0.11†
Talc poudrage	0.67	0.22 (2)	0.00	0.11†
Tunneled pleural catheter	0.00	0.51 (4), 0.49 (0)	0.10‡	0.00
Rapid Pleurodesis Protocol	0.70	0.22 (2), 0.08 (0)	0.00	0.00

*Per each thoracentesis procedure. Chiefly, pneumothorax requiring chest tube placement.

†Chiefly, previously undetected lung entrapment warranting tunneled pleural catheter placement.

‡Chiefly, infection requiring intravenous antibiotics.

equivalent, as discussed before. For RT, data quoted by Gordon et al's¹⁴ meta-analysis and the British Thoracic Society pleural disease guidelines was used.¹³ For RPP, the only identifiable source was Reddy et al's⁸ observational study (hence the sensitivity analysis as earlier mentioned). Events measured included transitions between major health states and complications that would lead to a hospitalization. Clinical outcome data, and by extension hospital LOS data, tends to be markedly skewed toward the right side.²³ As median is deemed a better measure of central tendency in skewed distributions, it was used instead of mean in all instances.²⁴

Health utility estimates were identical to those made in previously published cost-utility analyses for MPE.^{10,11} The major source of these estimates was a study ascertaining utilities for different health states related to metastatic non-small cell lung cancer.²⁵ Authors developed health state descriptions with high content validity before interviewing members of the general public to rate the utility of each state on the visual analog scale as well as by the standard gamble method. Borrowing from these, we assigned a utility of 0.473 (progressive cancer) to "unresolved effusion" and 0.599 (cancer, responding and fatigue) to "resolved effusion." For patients with an indwelling catheter (TPC), a utility of 0.58 was assigned. This was extrapolated from a meta-analysis of 6 studies assigning a utility to an indwelling peritoneal dialysis catheter using the EuroQol-5D questionnaire.²⁶

For patients switching across multiple health states, QALY estimates were calculated based on the approximate number of months spent in each state. QALYs were not discounted as the time horizon was within a 1-year timeframe.

Costs

Table 2 lists cost estimates for each intervention. All costs are expressed in 2014 US dollars.

Medicare reimbursements for hospitalization were obtained from published CMS data, and inflated from 2011 prices (the most recent ones available at the time of analysis) by 9.1% to estimate January 2014 values using the United States Consumer Price Index for urban customers of medical care.²⁷ Patients getting hospitalized for any intervention were assumed to incur the same cost as Medicare's average reimbursement per discharge for DRG 187 (pleural effusion with complications and comorbidities).^{28,29} For complications requiring hospitalization, DRG 186

(pleural effusion with major complications and comorbidities) was used.

Professional costs of procedures were estimated using Medicare CPT-based (Current Procedural Terminology) "National Payment Average" reimbursement amounts.³⁰ The facility price was used for outpatient procedures. This is relevant for outpatient, emergency, and ambulatory surgical facilities that are directly linked to hospitals (ie, "regulated space"). Nonfacility prices would be relevant for stand-alone centers without an accompanying hospital.³¹ The selection of billing codes was made after consultation with practicing interventional pulmonologists (H.L., L.Y., and D.J.F.-K.) as well as hospital billing professionals. Haas et al's³² review of billing and coding for pleural procedures was also consulted.

For ambulatory facilities accompanying hospitals, Medicare reimburses the facility using its Outpatient Prospective Payment System.³³ These costs were also included. CPT code 99213 (established patient, level III) was adopted for calculation of clinic costs in light of published literature as well as per clinical practice of authors (H.L., L.Y., and D.J.F.-K.).³⁴ Patients with RT would be expected to require 6 clinic visits during the course of this analytical horizon (a visit for each thoracentesis). After TS and TP, patients would not be expected to automatically require a follow-up visit. The TP and RPP groups would be expected to have no added clinic costs for up to 3 months in any case, as Medicare reimbursement for thoracoscopy with pleurodesis covers a 90-day global period. Patients with TPC placement would require an additional clinic visit for catheter removal if pleurodesis occurs. Clinic visits for suture removal were not accounted for, assuming absorbable sutures get used (although it may be noted that some centers reportedly use such a visit for additional purposes, eg, training). Medicare does not separately reimburse expenses related to equipment and supplies, and as such, those are included in the overall CPT-based reimbursement. Furthermore, Medicare reimburses home nursing visits in the form of 60-day packages. This payment is inclusive of all home health needs and supplies of care. Thus, costs of drainage bottles associated with an indwelling TPC were not separately included unless the patient/patient's caregiver was assumed to perform catheter care instead of a visiting nurse. The latter case was assumed to occur in 50% of cases.

Out-of-pocket non-health care costs (eg, transportation costs) and loss-of-productivity

TABLE 2. Cost Estimates for Each Intervention

Intervention Total Cost (2014 US \$)	Item	Item Costs (2014 US \$)	Description
Repeated thoracentesis \$3924 (over 6 mo)	Facility cost (hospital)	NA	Outpatient procedure
	Professional cost (procedure)	\$117	CPT code 32555, facility price (monthly)
	Professional cost (clinic)	\$52	CPT 99213 (monthly)
	Facility cost (clinic)	\$485	Medicare outpatient facility rate for CPT 32555 (monthly)
Talc slurry \$6159	Facility cost (hospital)	\$5905	DRG 187 (2011 values inflation-adjusted to 2014)
	Professional cost (procedure)	\$254	CPT codes 32557 + 32560, facility price
	Professional cost (clinic)	NA	Inpatient procedure
	Facility cost (clinic)	NA	Inpatient procedure
Talc poudrage \$6594	Facility cost (hospital)	\$5905	DRG 187; average reimbursement per discharge (2011 values adjusted per health care Consumer Price Index)
	Professional cost (procedure)	\$689	CPT code 32650, facility price
	Professional cost (clinic)	NA	Inpatient procedure
	Facility cost (clinic)	NA	Inpatient procedure
Tunneled pleural catheter 2 mo: \$4987 4 mo: \$6505 6 mo: \$7314	Facility cost (hospital)	NA	Outpatient procedure
	Professional cost (procedure)	\$291	CPT codes 32550 + 75989-26, facility price
	Facility cost (clinic)	\$2417	CPT code 32550
	Professional cost (clinic)—removal	\$172	CPT code 32552, facility price
	Facility cost (clinic)—removal	\$485	CPT code 32552
	Professional cost (clinic)*	\$52	CPT 99213 (2 visits if TPC removed)
	Visiting home nurse—60-day package	\$2100	CPT 99507: Contractor-priced. Inclusive of supplies of care (per 2 mo)
	Drainage kit	\$36	1 L drainage bottle along with dressing supplies; 3 drainage kits (\$108) consumed per week of self-care
Rapid Pleurodesis Protocol Success—TPC removed at 1 wk: \$7650 Stalemate—TPC stays: \$11,439	Facility cost (hospital)	\$5905	DRG 187; average reimbursement per discharge (2011 values adjusted per health care Consumer Price Index)
	Professional cost (procedure)	\$980	CPT codes 32650 + 75989-26 + 32550 (facility price)
	Professional cost (removal)	\$172	CPT code 32552, facility price
	Facility cost (clinic)	\$485	CPT code 32552 (TPC removal)
	Professional cost (clinic)	NA	Clinic visit for TPC removal covered under a 90-day global coverage period courtesy TP
Hospitalization due to complication \$9034	Facility cost (hospital)	\$9034	DRG 186; average reimbursement per discharge (2011 values adjusted per health care Consumer Price Index)

*Clinic visit for suture removal not factored in, as use of absorbable sutures assumed.
NA indicates not available; TPC, tunneled pleural catheter; TP, talc poudrage.

costs were not factored into this analysis, as those would not be relevant to the chosen (payer) perspective. Costs were not discounted as the time horizon was within a 1-year timeframe.

Willingness to Pay (WTP)

In the past, the “cost per QALY gained” threshold has been \$50,000/QALY, based on the notion that QALYs gained by patients from

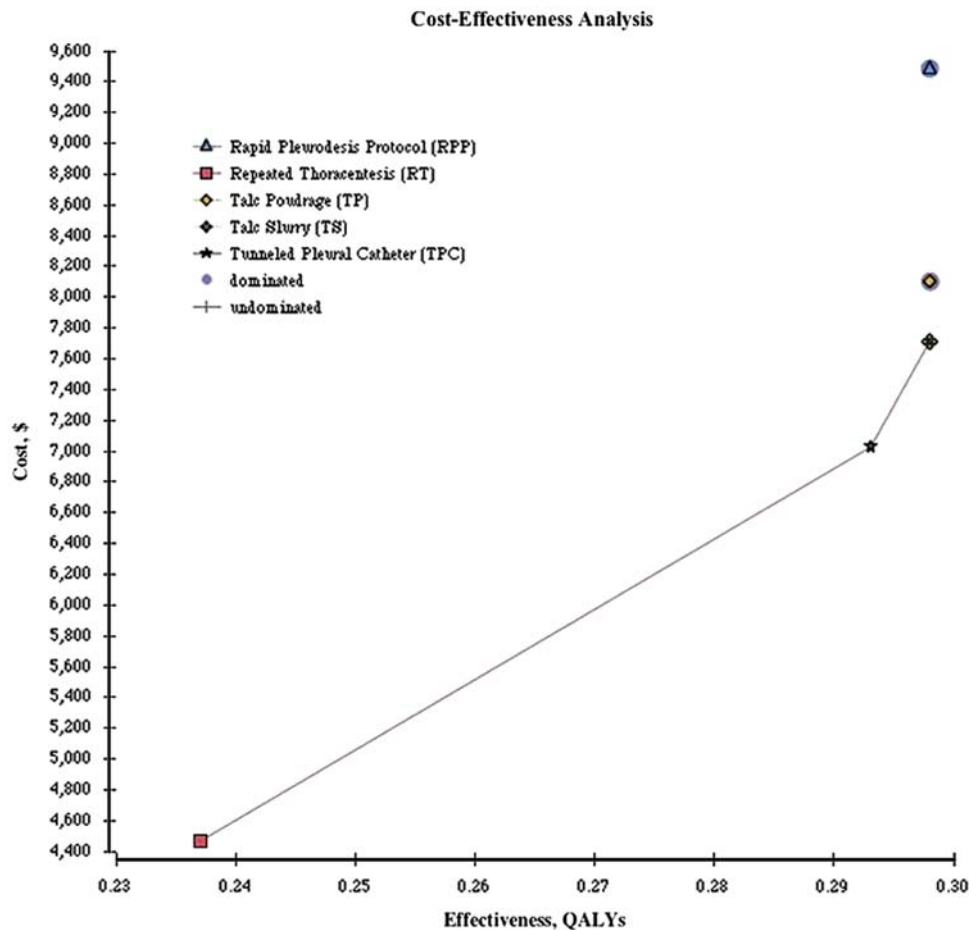


FIGURE 1. Base-case utility analysis. TP and RPP are dominated by TS. TPC, in turn, exerts extended dominance over TS. RPP indicates rapid pleurodesis protocol; TP, talc poudrage; TPC, tunneled pleural catheter; TS, talc slurry. *a+*

renal replacement therapy are worth the cost borne by society. Current studies frequently adopt \$100,000/QALY in light of a more recent analysis (2009) that estimated dialysis' ICER as \$110,814/QALY when compared with no dialysis.³⁵ Therefore, \$100,000/QALY was adopted as the WTP threshold in this study.

RESULTS

Figure 1 represents the base-case analysis results. TPC is more costly but more effective than RT. TS is more expensive and somewhat more effective than TPC but is associated with relatively greater cost per unit gain in QALY (steeper gradient). TPC and TS exhibit dominance over the

TABLE 3. Base-case Utility Analysis					
Intervention	Cost (\$)	Incremental Cost (\$)	Effectiveness (QALYs)	Incremental Effectiveness (QALYs)	ICER
Repeated thoracentesis	\$4466	—	0.24	—	—
Tunneled pleural catheter	\$7031	\$2565	0.29	0.06	\$45,747/QALY
Talc slurry	\$7714	\$684	0.30	0.00	\$137,492/QALY
Talc poudrage	\$8102	(Dominated)	0.30	(Dominated)	(Dominated)
Rapid Pleurodesis Protocol	\$9485	(Dominated)	0.30	(Dominated)	(Dominated)

ICER indicates incremental cost-effectiveness ratio.

TABLE 4. Multivariate (Nonrandom) Sensitivity Analysis

Permanent RPP Success*	Temporary RPP Success†	Intervention	Cost (\$)	Incremental Cost (\$)	Effectiveness (QALYs)	Incremental Effectiveness (QALYs)	ICER
0.70	0.22	Repeated thoracentesis	\$4466	—	0.24	—	—
		Tunneled pleural catheter	\$7031	\$2565	0.29	0.06	\$45,747/QALY
		Talc slurry	\$7714	\$684	0.30	0.00	\$137,492/QALY
		Talc poudrage	\$8102	(Dominated)	0.30	(Dominated)	(Dominated)
		Rapid Pleurodesis Protocol	\$9485	(Dominated)	0.30	(Dominated)	(Dominated)
0.75	0.17	Repeated thoracentesis	\$4466	—	0.24	—	—
		Tunneled pleural catheter	\$7031	\$2565	0.29	0.06	\$45,747/QALY
		Talc slurry	\$7714	\$684	0.30	0.00	\$137,492/QALY
		Talc poudrage	\$8102	(Dominated)	0.30	(Dominated)	(Dominated)
		Rapid Pleurodesis Protocol	\$9153	\$1439	0.30	0.00	\$6,073,456/QALY
0.80	0.12	Repeated thoracentesis	\$4466	—	0.24	—	—
		Tunneled pleural catheter	\$7031	\$2565	0.29	0.06	\$45,747/QALY
		Talc slurry	\$7714	\$684	0.30	0.00	\$137,492/QALY
		Talc poudrage	\$8102	(Dominated)	0.30	(Dominated)	(Dominated)
		Rapid Pleurodesis Protocol	\$8821	\$1107	0.30	0.00	\$2,169,065/QALY
0.85	0.07	Repeated thoracentesis	\$4466	—	0.24	—	—
		Tunneled pleural catheter	\$7031	\$2565	0.29	0.06	\$45,747/QALY
		Talc slurry	\$7714	\$684	0.30	0.00	\$137,492/QALY
		Talc poudrage	\$8102	(Dominated)	0.30	(Dominated)	(Dominated)
		Rapid Pleurodesis Protocol	\$8490	\$775	0.30	0.00	\$989,131/QALY

*Lasting pleurodesis achieved with RPP.

†Pleurodesis achieved initially, but effusion recurs at 2 months.

ICER indicates incremental cost-effectiveness ratio.

other 2 competing alternatives (TP and RPP), which offer little or no additional effectiveness despite higher costs.

Table 3 provides the base-case analysis results in detail. As compared with RT, TPC provides 0.06 additional QALYs and costs approximately \$2500 more, representing an ICER of \$45,747/QALY. This is within the WTP threshold of \$100,000/QALY, making TPC a preferable alternative to RT.

RT is the least expensive option, being approximately \$2500 less expensive than the next alternative. The other 4 incur costs within a \$2500 range from each other, although the least expensive among those (TPC) incurs only one-

third of the cost incurred by the most expensive one (RPP).

Table 4 provides results of multivariate (nonrandom) sensitivity analysis. As described in the Methods section, the probability of permanent RPP success was increased in 5% increments from 70% up to 85%. Although it was undominated in the 75% to 85% range, it never became a cost-effective strategy in the presence of other alternatives to RT with an ICER around or over \$1,000,000/QALY. Assuming an absence of other alternatives, however, the ICER of RPP over RT at a permanent success estimate of 85% would be \$65,091/QALY.

DISCUSSION

This cost-utility analysis of 5 therapeutic approaches to MPE shows that, for an analytical horizon of 6 months, TPC is the most cost-effective alternative to RT. This result is not sensitive to variation in the presumed efficacy of RPP.

The emergence of TPC as the most cost-effective intervention primarily results from its lower costs as compared with 3 of the other interventions (Tables 2, 3). QALY estimates were highly similar for all interventions except RT, which has the lowest QALYs.

We adopted an analytical horizon of 6 months. As previously described, most data points to an average survival of 6 months in patients with symptomatic MPE. For conditions whose associated costs and health impact do not cease until death, the commonly used approach is to use the average length of survival as the analytical horizon.²¹ Using the mean survival of all-comers with MPE as the time horizon also makes the results more meaningful from a third-party payer/health policy perspective. Furthermore, Puri et al's¹¹ analysis reported contrasting results using analytical horizons of 3 months and 12 months, making it particularly valuable to explore a 6-month horizon.

Our results add valuable insight to Puri et al's¹¹ findings, wherein TPC was the dominant strategy for an analytical horizon of 3 months but TS was more cost-effective for a 1-year horizon. In contrast to our findings, Olden and colleagues reported that TS dominated TPC at the 6-month horizon. One difference from our study was that they used a comparatively simplistic model that did not account for the need of a second intervention in case the first one failed.¹⁰ Both Puri and colleagues and Olden and colleagues used different efficacy estimates from ours due to nonavailability of currently available data at that time. However, these differences were too small to be majorly accountable for our study's findings. Neither Puri and colleagues nor Olden and colleagues provided sufficient details of their cost estimates as recommended in the CHEC-list guidelines.³⁶ A Dutch study, which estimated TPC costs from a health care facility's perspective, also showed that costs were similar to the hospitalization-driven costs associated with talc pleurodesis.³⁷ That study did not directly estimate TS or TP costs, however, and also did not estimate outcomes. No prior study evaluated RPP.

Penz et al³⁸ performed a post hoc cost estimation of TPC and TS using TIME2 trial participant data and nonparametric bootstrapping as a resampling strategy. Unlike our study, they did not perform cost-effectiveness analysis by contrasting costs with effectiveness data. After making inflation adjustment to 2013 values using the UK Consumer Price Index, they performed currency conversion to US dollars. They also used the third-party payer perspective (using the UK National Health Service data for the most part). However, their base-case cost estimates for TPC and TS were considerably lower than ours (\$4993 and \$4581, as opposed to \$7031 and \$7714, respectively). This is not surprising, given that health care expenditures in the United States are considerably greater than in the United Kingdom and other OECD countries.³⁹ Moreover, they made cost calculations relevant to the UK third-party payer, which are considerably different from those relevant to the US third-party payer (in terms of not only absolute values but also the actual items carrying a cost). Notably, the slightly greater cost of TPC relative to TS in their base-case analysis was reversed to match our pattern after certain costs not relevant to the US third-party payer were excluded during sensitivity analysis. Penz and colleagues's analysis also included significant hospitalization costs associated with TPC that arose not from the TPC itself but because 35% of patients were already hospitalized at the time of study enrollment.

Health utility values assigned to each outcome were the same as those assigned in Puri and colleagues and Olden and colleagues. Apart from the relevant studies' methodological soundness, this was done because it ensured that a reliable comparison was possible across all 3 studies based purely on revised estimation of costs and efficacies.

Our study attempts to perform much more thorough cost estimation than what has previously been published. The same is attempted for QALY calculations; whereas Puri and colleagues generally multiplied a single health utility value by the life expectancy to generate QALYs associated with each outcome arm, our QALY calculations accounted for subjects moving across multiple health states and assigned a separate utility value for any state lasting at least 1 month. Our analysis gave careful consideration to study quality (preferring RCTs) and time since publication (assuming no constancy of effect and preferring the most recent data) when assigning probability

estimates. Several important studies have been published since the last cost-effectiveness analysis on this subject, most notably the TIME2 trial, and also the Reddy et al⁸ study on the novel RPP.⁶ Unlike Olden and colleagues but similar to Puri and colleagues, our decision tree model acknowledged the possibility of requiring second-line interventions as well (Appendix A, Supplemental Digital Content 1, <http://links.lww.com/LBR/A120>). This provided for a more complete and accurate cost-utility analysis.

There is no reason to doubt that the common practice of prescribing RT for patients with a life expectancy of 1 month or less would still be appropriate.³ As RT is the least expensive intervention available, our study's results would not leave this subset of patients vulnerable to denial of coverage from third-party payers.

This study adds to the breadth and depth of information available on the cost-utility of various palliative interventions for MPE. The policy implications are obvious: insurance coverage should be readily available for TPC placement to all MPE patients—even those that do not show evidence of a nonexpandable lung. That said, as the 3 published cost-effectiveness analyses together suggest that an intervention's cost-effectiveness might vary with the analytical horizon, further investigation is warranted into whether the preferred intervention should vary based on the average survival time of a given type of malignancy (eg, breast cancer vs. sarcoma). Although it may not alter our conclusion, our results are limited by the fact that cost estimates would need readjustment for certain other situations. These include stand-alone ambulatory practices or rural areas where Medicare's National Payment Average would not apply perfectly, and those payers that reimburse hospitalizations based on LOS. Another issue common to procedural interventions is that of expertise, which may not be consistent across providers and institutions. As this analysis used outcome probabilities based largely on the TIME2 RCT data, it would be most representative of academic, high volume, and/or hospital-based facilities.

One limitation inherent to the decision tree model is that, if the initial approach does not succeed and multiple second-line options are available, the analysis results would vary based on which path(s) is undertaken. Fortunately, all competing alternatives in this study were estimated to have low failure rates except for both TS and TP (approximately 10% in each case). In

these cases, however, it could be argued that there is an overwhelmingly more appropriate next step on offer. As talc pleurodesis failure is commonly attributable to previously unsuspected lung entrapment, a TPC placement can be justified as the logical next step.³ In our model, we adopted the most appropriate second-line approaches based on a review of literature as well as discussion among the authors.^{2,3} What our analysis (and previous analyses) cannot account for are the need to first rule out lung entrapment (for which a therapeutic thoracentesis would be most appropriate), and the need to definitively diagnose MPE (for which either a thoracentesis or thoracoscopic pleural biopsy would be most appropriate). It is noteworthy that our model assumes an initial outpatient presentation and would therefore not be entirely applicable to situations and practices where the initial workup or management might be carried out within an existing hospitalization. This would have an obvious impact on cost calculations. In contrast, as we have chosen a well-defined starting point in our decision tree model, our results are all the more valid for the clinical scenario of greatest relevance: patients referred to an interventional pulmonary/pleural disease clinic for management of already diagnosed, symptomatic MPE. Surgical interventions such as decortication and VATS pleurodesis, or experimental techniques such as intrapleural chemotherapy or gene therapy, were not included in our analysis.²

Our model assumes the patient population to be similar to what was represented in the TIME2 and Reddy and colleagues studies. Both studies had broadly similar enrollment criteria: consenting adults with established symptomatic MPE with the exclusion of those with limited expected survival (defined as <1 month by Reddy and colleagues and <3 months by the TIME2 investigators). Another important consideration is that, as a cost-effectiveness analysis is based on aggregate data reflecting population averages, it is not directly applicable to the individual patient, where additional factors such as comorbidities, accessibility, and—most of all—patient preferences, come into play. Even with respect to costs, the patient perspective could be considerably distinct from the third-party payer perspective. For example, while RT is easily the most economical option from the latter perspective, it could entail greater patient-related costs from more frequent trips to the clinic or emergency room and greater loss of productivity (days of work missed by caregivers or patients themselves). Further, it is

difficult to place a value on various “valuable” features of an intervention, for example, ethical considerations (is the greater short-term discomfort of RTs or a severe talc reaction ethically justifiable, even if such an approach were more cost-effective? Or the hassle and/or embarrassment of having to constantly deal with an indwelling pleural catheter?). Hence, a cost-effectiveness analysis, which basically adds cost considerations to conventional comparative effectiveness research, is a necessary consideration during policy making but cannot guide clinical decisions at the bedside.²¹

CONCLUSIONS

For patients with MPEs and an expected survival of 6 months, TPC is the most cost-effective alternative to RT for pleural palliation. Chemical pleurodesis by TS is the next most cost-effective option. Larger studies are required to examine the long-term efficacy of RPP, as well as to investigate if the protocol can be implemented on a purely outpatient basis. It remains to be seen whether the results of such investigation would make RPP a viable alternative to TPC.

ACKNOWLEDGMENTS

The authors gratefully acknowledge the assistance provided by several billing professionals at one of our institutions, including Renea Olson, Dawn Hohl, and Sue Snyder (no compensation was received for these efforts).

REFERENCES

- Villaneuva A. Management of malignant pleural effusions. In: Ernst A, Herth FJF, eds. *Principles and Practice of Interventional Pulmonology*. New York, NY: Springer; 2013:665–674.
- Thomas JM, Musani AI. Malignant pleural effusions: a review. *Clin Chest Med*. 2013;34:459–471.
- Roberts ME, Neville E, Berrisford RG, et al. Management of a malignant pleural effusion: British Thoracic Society Pleural Disease Guideline 2010. *Thorax*. 2010;65(suppl 2):ii32–ii40.
- Rodriguez-Panadero F, Borderas Naranjo F, Lopez Mejias J. Pleural metastatic tumours and effusions. Frequency and pathogenic mechanisms in a post-mortem series. *Eur Respir J*. 1989;2:366–369.
- Burrows CM, Mathews WC, Colt HG. Predicting survival in patients with recurrent symptomatic malignant pleural effusions: an assessment of the prognostic values of physiologic, morphologic, and quality of life measures of extent of disease. *Chest*. 2000;117:73–78.
- Davies HE, Mishra EK, Kahan BC, et al. Effect of an indwelling pleural catheter vs chest tube and talc pleurodesis for relieving dyspnea in patients with malignant pleural effusion: the TIME2 randomized controlled trial. *JAMA*. 2012;307:2383–2389.
- Clive AO, Kahan BC, Hooper CE, et al. Predicting survival in malignant pleural effusion: development and validation of the LENT prognostic score. *Thorax*. 2014;69:1098–1104.
- Reddy C, Ernst A, Lamb C, et al. Rapid pleurodesis for malignant pleural effusions: a pilot study. *Chest*. 2011;139:1419–1423.
- Putnam JB Jr, Light RW, Rodriguez RM, et al. A randomized comparison of indwelling pleural catheter and doxycycline pleurodesis in the management of malignant pleural effusions. *Cancer*. 1999;86:1992–1999.
- Olden AM, Holloway R. Treatment of malignant pleural effusion: PleuRx catheter or talc pleurodesis? A cost-effectiveness analysis. *J Palliat Med*. 2010;13:59–65.
- Puri V, Pyrdeck TL, Crabtree TD, et al. Treatment of malignant pleural effusion: a cost-effectiveness analysis. *Ann Thorac Surg*. 2012;94:374–379. discussion 79–80.
- Light RW. *Pleural Diseases*, 6th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2013.
- Havelock T, Teoh R, Laws D, et al. Pleural procedures and thoracic ultrasound: British Thoracic Society Pleural Disease Guideline 2010. *Thorax*. 2010;65(suppl 2):ii61–ii76.
- Gordon CE, Feller-Kopman D, Balk EM, et al. Pneumothorax following thoracentesis: a systematic review and meta-analysis. *Arch Intern Med*. 2010;170:332–339.
- Dresler CM, Olak J, Herndon JE II, et al. Phase III intergroup study of talc poudrage vs talc slurry sclerosis for malignant pleural effusion. *Chest*. 2005;127:909–915.
- Gonzalez AV, Bezwada V, Beamis JF Jr, et al. Lung injury following thorascopic talc insufflation: experience of a single North American center. *Chest*. 2010;137:1375–1381.
- Rinaldo JE, Owens GR, Rogers RM. Adult respiratory distress syndrome following intrapleural instillation of talc. *J Thorac Cardiovasc Surg*. 1983;85:523–526.
- Musani AI, Haas AR, Seijo L, et al. Outpatient management of malignant pleural effusions with small-bore, tunneled pleural catheters. *Respiration*. 2004;71:559–566.
- Tremblay A, Michaud G. Single-center experience with 250 tunnelled pleural catheter insertions for malignant pleural effusion. *Chest*. 2006;129:362–368.
- Fysh ET, Tremblay A, Feller-Kopman D, et al. Clinical outcomes of indwelling pleural catheter-related pleural infections: an international multicenter study. *Chest*. 2013;144:1597–1602.
- Muenig P. *Cost-Effectiveness Analysis in Health: A Practical Approach*, 2nd ed. San Francisco, CA: Jossey-Bass; 2008.
- Overview of the Medicare and Medicaid programs. *Health Care Financ Rev Stat Suppl*. 1999;1–348.
- Faddy M, Graves N, Pettitt A. Modeling length of stay in hospital and other right skewed data: comparison of phase-type, gamma and log-normal distributions. *Value Health*. 2009;12:309–314.
- Gordis L. *Epidemiology*, 4th ed. Philadelphia, PA: Saunders; 2009.
- Nafees B, Stafford M, Gavriel S, et al. Health state utilities for non small cell lung cancer. *Health Qual Life Outcomes*. 2008;6:84.
- Liem YS, Bosch JL, Hunink MG. Preference-based quality of life of patients on renal replacement therapy:

- a systematic review and meta-analysis. *Value Health*. 2008;11:733–741.
27. Consumer Price Index. Secondary Consumer Price Index. Available at: <http://www.bls.gov/cpi/>. Accessed April 20, 2014.
 28. List of Diagnosis-related Groups (DRGs), FY 2011. Secondary List of Diagnosis-related Groups (DRGs), FY 2011. Available at: <http://www.cms.gov/Research-Statistics-Data-and-Systems/Statistics-Trends-and-Reports/MedicareFeeforSvcPartsAB/Downloads/DRGdesc11.pdf>. Accessed April 20, 2014.
 29. 2011 Short Stay Inpatient by DRG [PDF, 242KB]. Secondary 2011 Short Stay Inpatient by DRG [PDF, 242KB]. Available at: <http://www.cms.gov/Research-Statistics-Data-and-Systems/Statistics-Trends-and-Reports/MedicareFeeforSvcPartsAB/Downloads/DRG11.pdf>. Accessed April 20, 2014.
 30. Physician Fee Schedule Search. Secondary Physician Fee Schedule Search. Available at: <http://www.cms.gov/apps/physician-fee-schedule/search/search-criteria.aspx>. Accessed April 20, 2014.
 31. How to Use The Searchable Medicare Physician Fee Schedule (MPFS). Secondary How to Use The Searchable Medicare Physician Fee Schedule (MPFS). Available at: http://www.cms.gov/Outreach-and-Education/Medicare-Learning-Network-MLN/MLNProducts/downloads/How_to_MPFS_Booklet_ICN901344.pdf. Accessed April 20, 2014.
 32. Haas AR, Sterman DH, Musani AI. Malignant pleural effusions: management options with consideration of coding, billing, and a decision approach. *Chest*. 2007;132:1036–1041.
 33. Hospital Outpatient Prospective Payment System. Secondary Hospital Outpatient Prospective Payment System. Available at: <http://www.cms.gov/Outreach-and-Education/Medicare-Learning-Network-MLN/MLNProducts/downloads/HospitalOutpaysysfctsht.pdf>. Accessed April 20, 2014.
 34. Waller TA. Level-II vs. level-III visits: cracking the codes. *Fam Pract Manag*. 2007;14:21–25.
 35. Lee CP, Chertow GM, Zenios SA. An empiric estimate of the value of life: updating the renal dialysis cost-effectiveness standard. *Value Health*. 2009;12:80–87.
 36. Evers S, Goossens M, de Vet H, et al. Criteria list for assessment of methodological quality of economic evaluations: Consensus on Health Economic Criteria. *Int J Technol Assess Health Care*. 2005;21:240–245.
 37. Boshuizen RC, Onderwater S, Burgers SJ, et al. The use of indwelling pleural catheters for the management of malignant pleural effusion—direct costs in a Dutch hospital. *Respiration*. 2013;86:224–228.
 38. Penz ED, Mishra EK, Davies HE, et al. Comparing cost of indwelling pleural catheter vs talc pleurodesis for malignant pleural effusion. *Chest*. 2014;146:991–1000.
 39. Lorenzoni L, Belloni A, Sassi F. Health-care expenditure and health policy in the USA versus other high-spending OECD countries. *Lancet*. 2014;384:83–92.